

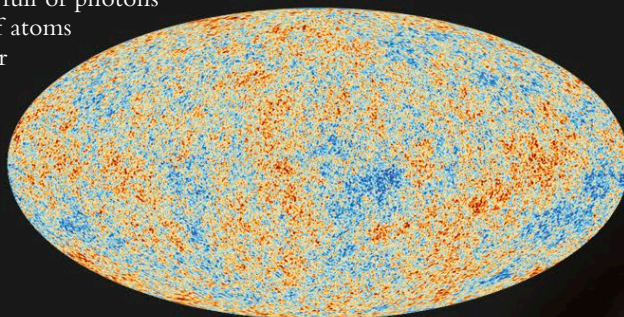
OUT OF THE DARKNESS

28–31	Matter
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THE PERIOD FROM THE BIRTH of atoms 380,000 years after the Big Bang to the ignition of the first stars hundreds of millions of years later is known as the “dark ages” of the universe. What happened in this era, and the subsequent “cosmic renaissance” as starlight filled the universe, is an intricate puzzle. Astronomers are solving it by analyzing the relic radiation of the Big Bang and using the world’s most powerful telescopes to peer to the edges of the universe.

THE AFTERMATH OF THE BIG BANG

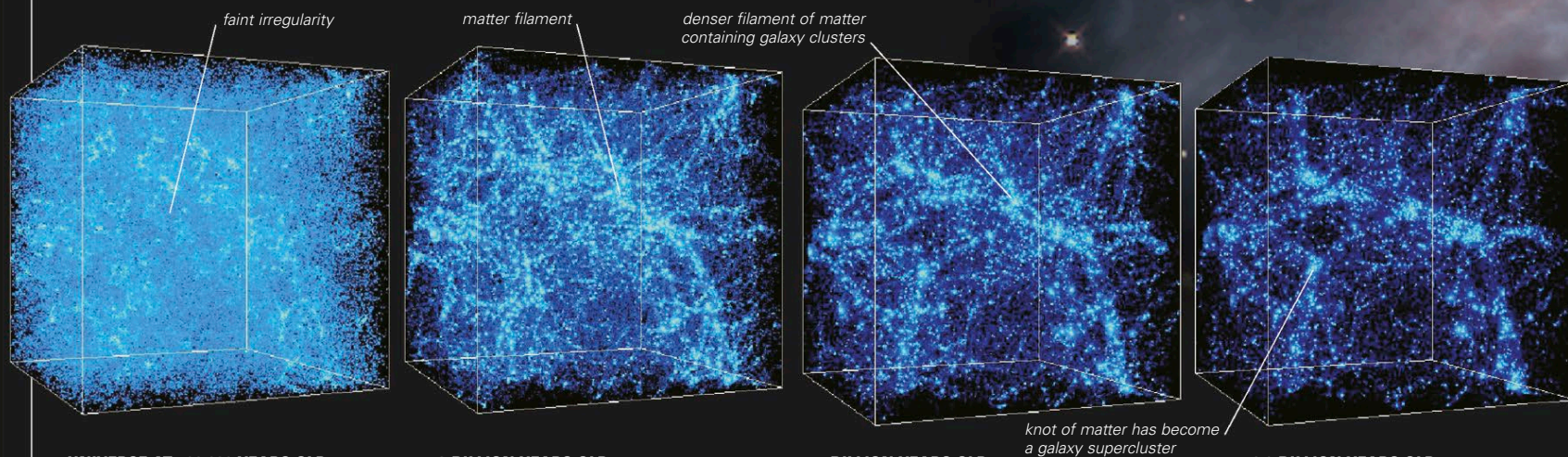
At an age of 400,000 years, the universe was full of photons of radiation streaming in all directions and of atoms of hydrogen and helium, neutrinos, and other dark matter. Although it was still hot, at 5,400°F (3,000°C), and full of radiation, astronomers see no light if they try to peer back to that moment. The reason is that as the universe has expanded, it has stretched the wavelengths of radiation by a factor of a thousand. The photons reach Earth not as visible light, but as low-energy photons of cosmic microwave background radiation (CMBR). Their wavelength, once characteristic of the fireball of the universe, is now that of a cold object with a temperature of –454°F (–270°C)—only 5°F (3°C) above absolute zero.



INFANT UNIVERSE
This Planck Observatory image is an all-sky picture of tiny fluctuations in the temperature of the CMBR, which relate to early irregularities in matter density. In effect, it is a snapshot of the infant universe.

THE DARK AGES

Earth will never receive visible light from the period before the first stars ignited, a few hundred million years after the Big Bang, but cosmologists can reconstruct what happened during that time using other data, such as those of the CMBR. The CMBR reveals tiny fluctuations in the density of matter at the time the first atoms formed. Cosmologists think that gravity working on these ripples caused the matter to begin forming into clumps and strands. These irregularities in the initial cloud of matter probably laid the framework of present-day large-scale objects, such as galaxy superclusters (see pp.336–337). The development of such structures over billions of years has been simulated with computers. These simulations rely on assumptions about the density and properties of matter, including dark matter, in the infant universe, as well as the influence of dark energy (a force opposing gravity, see p.58). Some simulations closely resemble the distribution of matter seen in the universe today.



UNIVERSE AT 500,000 YEARS OLD
This computer simulation of the development of structure in the universe starts with matter almost uniformly dispersed in a cube that is 140 million light-years high, wide, and deep.

1.3 BILLION YEARS OLD
A billion years later, considerable clumping and filament formation has occurred. To compensate for the cosmic expansion since the previous stage, the cube has been scaled to size.

5 BILLION YEARS OLD
A further 4 billion years later, and (again, after rescaling) the matter has condensed into some intricate filamentous structures interspersed with sizable bubbles or voids of empty space.

13.8 BILLION YEARS OLD
The matter distribution in the simulation now resembles the kind of galaxy-supercluster structure seen in the local universe (within a few billion light-years).